META-ANALYTIC EVALUATION OF VOLLEYBALL METRICS

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INTRODUCTION

- **Prelude**:
  - Sports analytics now produces *hundreds* of player metrics, to attempt to capture skill, context, and chance variability.
  - Many metrics are redundant or noisy, with contributions often unknown, creating clutter and lack of transparency.
  - Decision makers need to know which metrics are reliable, stable and meaningful
- **Goal:** Apply meta-analytics (Franks et al, 2017) to sports statistics

Defining Meta-Metrics (Franks et al, 2017)

- We want to quantify *sources of variability* and how they relate across the dimensions of metrics (**m**), players (**p**), and season (**s**):
- We assume that each player metric is composed of a combination of:
  - *intrinsic player skill (*what we care about)
  - *context* (e.g. the influence of teammates)
  - *chance* (i.e sampling variability)
- We want to understand how well certain metrics can measure, control for, and average across these sources of variation, and we categorize these sources as either *signal* or *noise*
  - *signal:* variation that is a key input into a decision process
  - *noise:* variation that we don't explain due to complexity or ambiguity

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Defining Meta-Metrics (Franks et al, 2017)

- A metric's treatment of variation must align with the decision that it's being used for
  - In *credit assignment* problems, we aren't concerned about the repeatability of a performance
  - In *player acquisition* problems, we are interested in the repeatability of a performance
- We consider three meta-metrics:
  - ***discrimination***
  - ***stability***
  - ***independence***

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Notation / STAT 2xx Review

- X_{spm} is the value of metric m for player p in season s sampled from a random variable X
  - Eg: $X_{\{2025, Shohei\ Ohtani, Batting\ Average\}} = .282$
- (Sample) Mean/average: $E[X] = \frac{\Sigma_i x_i}{n}$
  - $E_{spm}[X] = E[X|S = s, P = p, M = m]$
- (Sample) Variance: $V[X] = \frac{\Sigma(x - \bar{x})^2}{n-1}$
  - $V_{spm}[X] = Var[X|S = s, P = p, M = m]$
- Example:
  - $E_{sm}\left[V_{spm}[X]\right] = \frac{\Sigma_p Var[X_{spm}]}{N_{sm}}$ is the average over all players of the intrinsic variability in X_{spm} for metric m during season s

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Intuition

- R-squared: $R^2 = 1 - \frac{SS_{RES}}{SS_{TOT}} = 1 - \frac{\Sigma_i (y_i - \hat{y}_i)^2}{\Sigma_i (y_i - \bar{y}_i)^2}$
- R-squared measures the proportion of variance in a response variable that a (set of) dependent variables are able to predict. A higher value indicates a better model fit, while a smaller value indicates a weaker fit.
- Each meta-metric is defined in terms of isolating for the ratio of *signal* to *total variance* by subtracting the ratio of noise to total variance from 1.

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META-METRIC 1: Discrimination

- Does the metric differentiate between players?

$$D_{sm} = 1 - \frac{E_{sm}[V_{spm}[X]]}{V_{sm}[X]}$$

- Represents the proportion of average sampling variability to total between player variance in metric m that reflects true differences in player ability rather than sampling noise
- Example: Fielding% vs Fielding Run Value
  - Both are measuring defensive proficiency, but FRV is much more discriminative than Fielding%

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META-METRIC 2: Stability

- Does the metric measure the same thing over time?

$$S_m = 1 - \frac{E_m[V_{pm}[X] - V_{spm}[X]]}{V_m[X] - E_m[V_{spm}[X]]}$$

with $0 \leq S_m \leq 1$

- Quantifies how much of a player's variation in metric m remains consistent across seasons after accounting for random noise
- Example: Free Throw % vs 3-point %
  - Both measure shooting ability, but FT% is much more stable from year-to-year than 3PT%

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META-METRIC 3: Independence

- Does the metric provide new information?

$$Z_{sp} \sim MVN(0, C)$$

$$X_{spm} = F_m{}^{-1}[\Phi(Z_{spm})]$$

$$\mathcal{I}_{m\mathcal{M}} = \frac{Var\left[Z_{spm} \mid \{Z_{spq} : q \in \mathcal{M}\}\right]}{Var[Z_{spm}]}$$

$$= C_{m,m} - C_{m,\mathcal{M}} C^{-1}{}_{\mathcal{M},\mathcal{M}} C_{\mathcal{M},m}$$

- Measures how much unique information that metric m adds beyond the set of other metrics M, with metric values normalized to remove differences in units and scaling

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BACKGROUND

- **Motivation**: Volleyball is one of the fastest-growing sports in the world yet surprisingly lacks a systematic meta-analytical review of its various metrics comparable to the attention given to basketball, hockey, and soccer.
- **Application**: Metric selection for scouting, salaries, performance evaluation, lineup decisions, and metric development.
- **Considerations**: Practical significance, sampling variability, data availability, metric redundancy, and normalization across seasons and players.
- **Framework:** Metrics blend intrinsic skill, contextual influences, and change; meta-analytics separates these sources of variation into meaningful signal vs. noise.

Volleyball

- Volleyball is played by two teams that each occupy a 9x9 meter side of a court separated by a net that is 224 centimeters tall for women.
- A volleyball match is divided into segments called *sets*, where the first team to win 3 sets wins a match.
- A point is initiated by a serve from the team that won the most recent point, and teams alternate possessions by returning the ball over the net after performing a maximum of three consecutive contacts on their side.
- There can be 6 players on the court at any time, 3 in the front row and 3 in the back row.
- The team that receives the serve rotates one position clockwise if they win the point.

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Data Exposition

- There are several repositories of publicly available volleyball data that we considered:
  - **usportsR/usportspy**: player and team data from University Sports (U SPORTS)
  - **rvolleydata/pyvolleydata**: player and team statistics from LOVB Pro, MLV, and AU Pro Volleyball
  - **ncaavolleyballr**: player and team statistics from the NCAA Division I, II, and II volleyball leagues

METHODS: VOLLEYBALL METRICS

Metric	Definition	Calculation
Kills (K)	Attacks that scores a point	Counting statistic
Attack Errors (E)	Attack attempts resulting in opponent's point	Counting statistic
Total Attempts (TA)	All swings by an attacker, whether they score or not	Counting statistic
Kill Percentage (K%)	How often an attacker's swing becomes a kill	Kills / Total Attempts
Attack Efficiency (PCT)	How efficiently a player scores when attacking	(Kills – Attack Errors) / Total Attempts
Service Aces (SA)	Serves that land in and the other team cannot control	Counting statistic
Service Errors (SE)	Serves that go out, into the net, or are foot faults	Counting statistic
Serve Efficiency (S%)	Accuracy of service	(SA – SE) / (SA + SE)

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METHODS: VOLLEYBALL METRICS

Metric	Definition	Calculation
Assists (A)	Sets that lead directly to a kill by an attacker	Counting statistic
Ball-handling Error (BHE)	A violation called when a player fails to contact the ball cleanly, resulting in it being caught, thrown, held, or double touch.	Counting statistic
Receiving Error (RE)	Opponent gets service ace (unable to return/keep in play)	Counting statistic
Digs (D)	Defensive plays that successfully passes a ball that has been attacked by the opponent or after a block touch	Counting statistic
Block Solo (BS)	One blocker stops the opponent's attack and wins the point directly	Counting statistic
Block Assists (BA)	2+ blockers together stop the opponent's attack and win the point	Counting statistic
Total Blocks (TB)	Blocks that result in a point for the blocking team	BS + 0.5 * BA
Total Points (PTS)	Points earned by kills, aces, and blocks (does not count points earned by opponent's errors)	K + SA + Total Blocks

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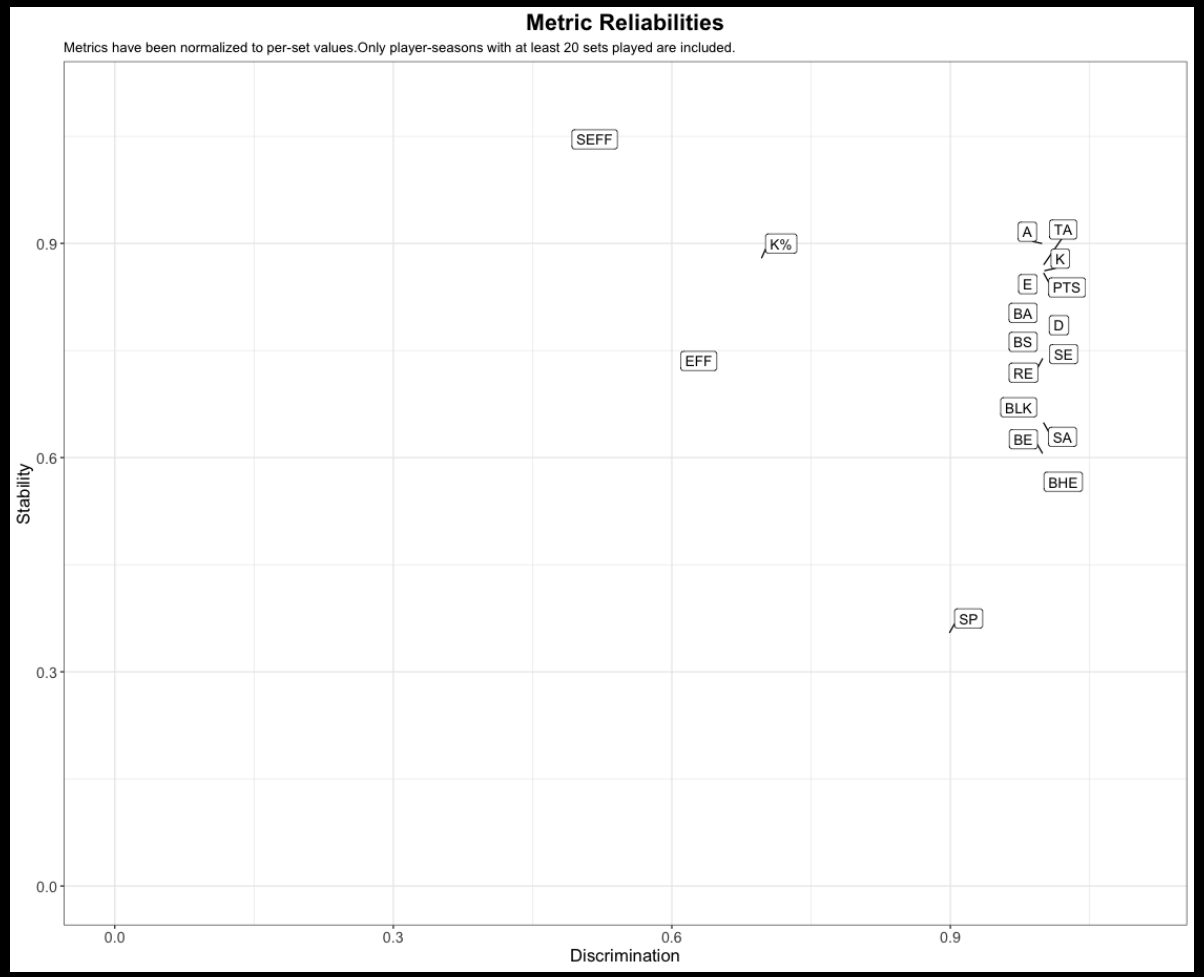
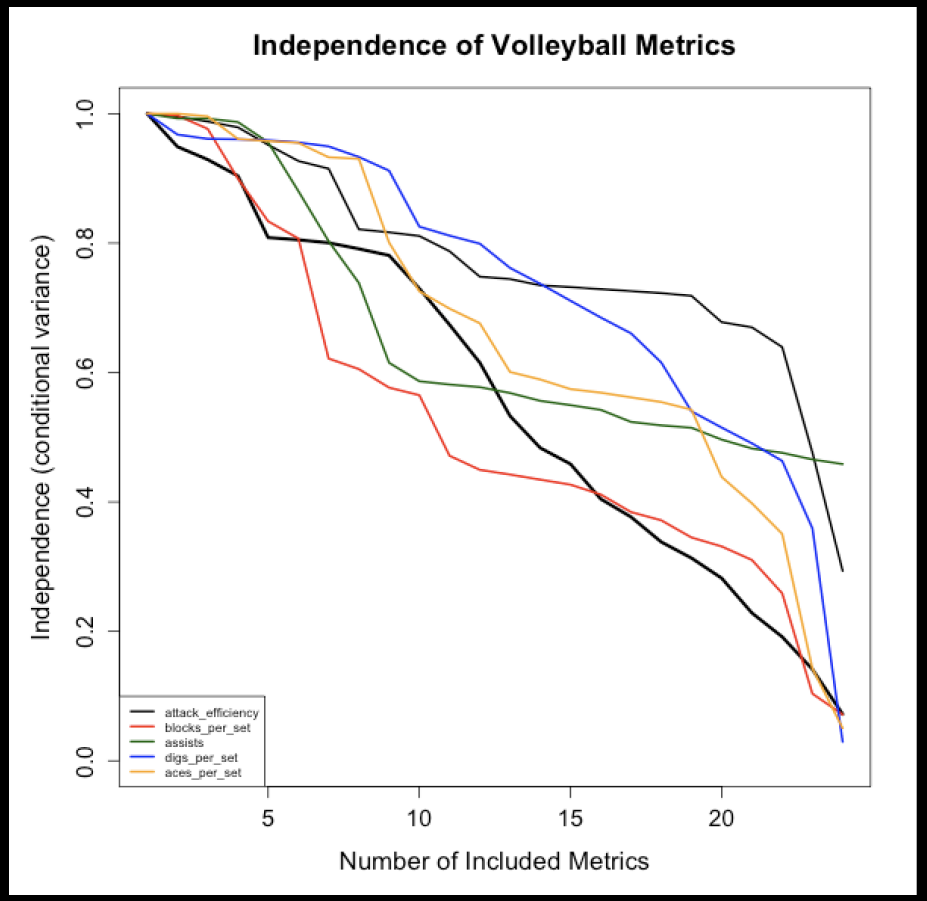
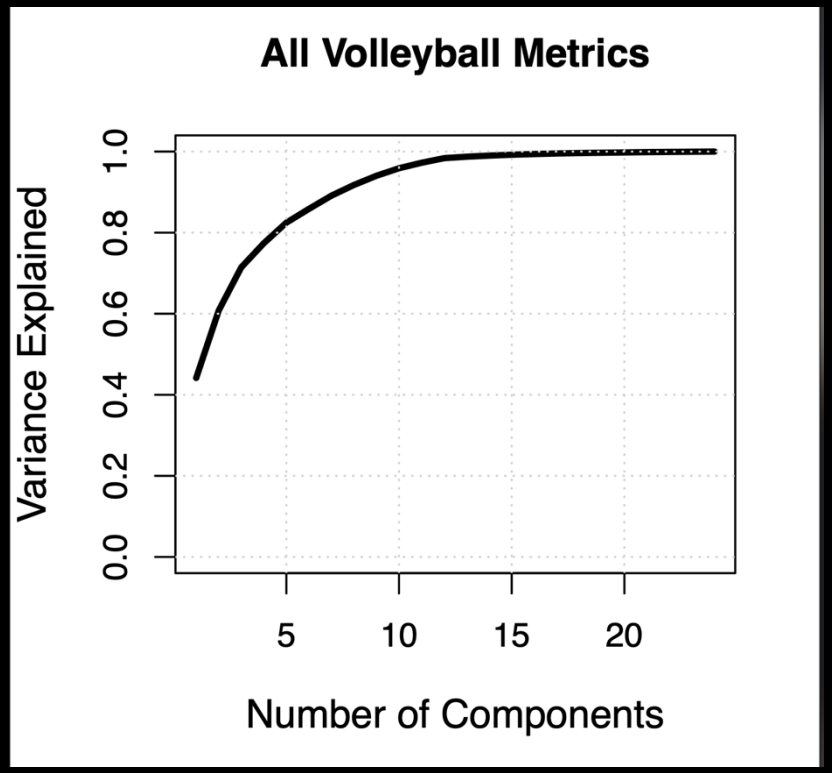
DISCUSSION: Stability & Discrimination

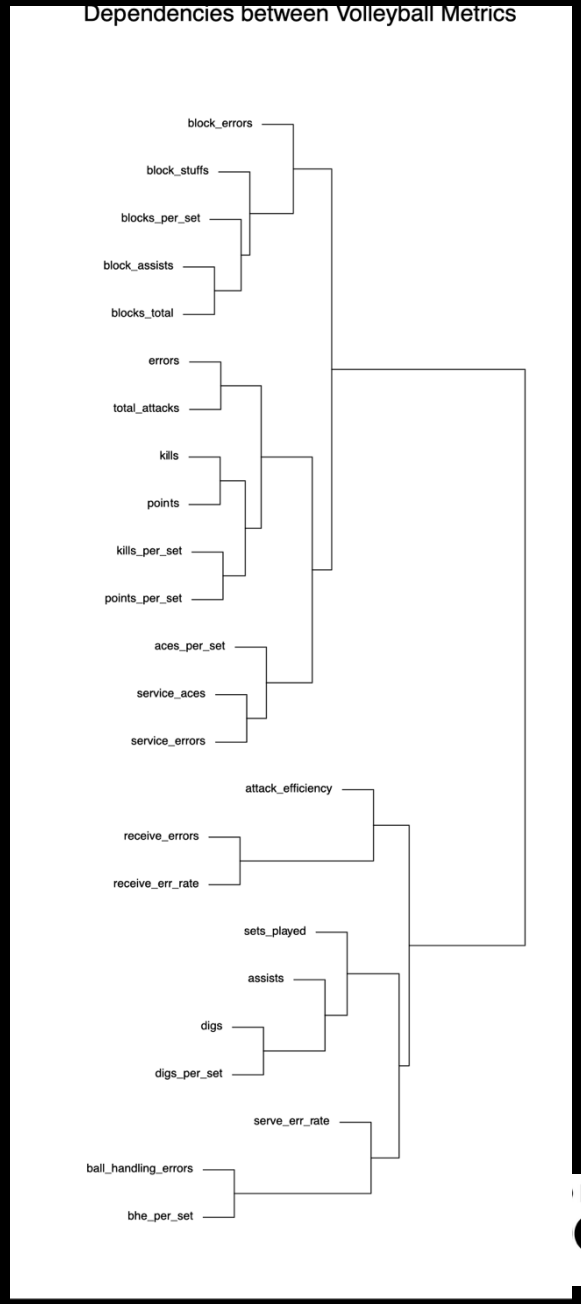


- Most volleyball metrics seem quite discriminatory, with varying levels of stability
- Serve efficiency (SEFF) is incredibly stable from year to year but lacks the discriminatory power of aces and service errors
- Attack efficiency is both less stable and less discriminative than just using kill%

DISCUSSION: Independence







CONCLUSION

- We can apply the meta-analytic framework to evaluate how well metrics capture player performance
- By separating skill, context, and chance, we can identify which metrics truly reflect meaningful differences between players
- Discrimination, stability, and independence together provide a structured way to compare metrics across seasons and players
- This approach helps teams and analysts focus on metrics that support clearer, more consistent player evaluation

FUTURE WORK

- Since the career length of a collegiate player is capped at 4-5 years and the professional women's leagues are relatively nascent, meta-metric calculations are more variable than those made from established professional leagues.
- This analysis focused on the coarsest level of volleyball statistics – boxscore data
  - Future extensions of this work could consider metrics derived from more granular sources of data, such as play-by-play (event-level) data and tracking data (data with spatial coordinates and a temporal dimension)
- There could be opportunity to create custom metrics that fill existing gaps in the volleyball meta-analytic literature using techniques such as Empirical Bayes.

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Thank you for your attention!

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